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# Looking Beyond the Window: Global-Local Aligned CLIP for Training-free Open-Vocabulary Semantic Segmentation

ByeongCheol Lee, Hyun Seok Seong, Sangeek Hyun, Gilhan Park, WonJun Moon, Jae-Pil Heo*
Sungkyunkwan University
{bc7817, gustjrdl95, hsi1032, a01152a, wjun0830, jaepilheo}@skku.edu

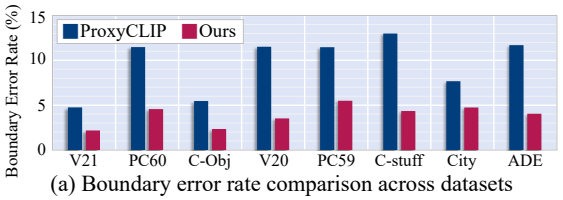
## Abstract

*A sliding-window inference strategy is commonly adopted in recent training-free open-vocabulary semantic segmentation methods to overcome limitation of the CLIP in processing high-resolution images. However, this approach introduces a new challenge: each window is processed independently, leading to semantic discrepancy across windows. To address this issue, we propose Global-Local Aligned CLIP (GLA-CLIP), a framework that facilitates comprehensive information exchange across windows. Rather than limiting attention to tokens within individual windows, GLA-CLIP extends key-value tokens to incorporate contextual cues from all windows. Nevertheless, we observe a window bias: outer-window tokens are less likely to be attended, since query features are produced through interactions within the inner window patches, thereby lacking semantic grounding beyond their local context. To mitigate this, we introduce a proxy anchor, constructed by aggregating tokens highly similar to the given query from all windows, which provides a unified semantic reference for measuring similarity across both inner- and outer-window patches. Furthermore, we propose a dynamic normalization scheme that adjusts attention strength according to object scale by dynamically scaling and thresholding the attention map to cope with small-object scenarios. Moreover, GLA-CLIP can be equipped on existing methods and broad their receptive field. Extensive experiments validate the effectiveness of GLA-CLIP in enhancing training-free open-vocabulary semantic segmentation performance. Codes are available at github.com/2btlFe/GLA-CLIP*

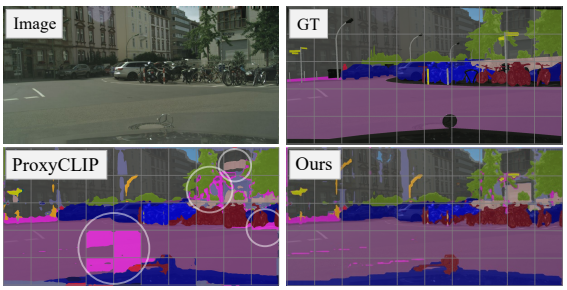
## 1. Introduction

Open-vocabulary semantic segmentation (OVSS) aims to assign pixel-level semantic labels from an open and potentially unbounded vocabulary, enabling models to generalize beyond a fixed set of training categories. Recent advances in vision-language models, such as CLIP [34], have made it possible to approach this task without additional training, by leveraging the image-text aligned embedding space. This has led to a growing interest in Training-Free OVSS [3, 18, 23, 24, 28, 38, 42, 50, 53], where segmentation is performed only with minimal modifications to the CLIP architecture or processing pipeline.

However, a fundamental limitation remains: CLIP is pretrained on low-resolution inputs (e.g., 224×224), which hinders its applicability to high-resolution segmentation tasks. To address this, recent works adopt a sliding-window strat-




(a) Boundary error rate comparison across datasets




(b) Grid artifacts from contextual discrepancy across windows

Figure 1. Comparison of segmentation consistency near window boundaries. (a) We evaluate segmentation inconsistency using the Boundary Error Rate (BER). BER is defined as the proportion of pixels near adjacent window boundaries where predicted labels differ despite identical ground-truth. ProxyCLIP yields high BER due to its lack of cross-window interaction, whereas BER is significantly reduced in ours by incorporating global context into attention process. More details in Appendix A. (b) ProxyCLIP exhibits grid artifacts (marked with white circles), caused by the limited receptive field within individual windows. In contrast, ours mitigates these artifacts by leveraging contextual information beyond local windows.

*Corresponding author.

egy [18, 22–24, 42, 53], where high-resolution images are divided into overlapping crops and processed independently. By extending precise text-aligned embeddings to high-resolution settings without compromising spatial details, this simple approach has led to a significant performance boost. However, we argue that this strategy still has a critical limitation: each window lacks access to the broader scene context, which leads to inconsistent predictions across windows.

One clear example of this limitation is illustrated in Fig. 1, where independently processed windows produce inconsistent predictions near window boundaries and introduce grid-like artifacts. Specifically, adjacent windows often assign different semantic labels to neighboring pixels along shared boundaries, even when those pixels belong to the same ground-truth class, resulting in visible discontinuities and structured artifacts. These issues highlight the absence of global context integration and motivate the need for a mechanism that enforces semantic consistency across windows.

To address this issue, we propose Global-Local Aligned CLIP (GLA-CLIP), a novel training-free framework that explicitly aligns local and global semantics. We begin by introducing a Key-Value Extension mechanism that allows each query to attend to an unbounded set of key-value tokens across the entire image. This design enables broader context integration, thereby mitigating the semantic discrepancy across independently processed windows. However, we observe that even with globally extended key-value tokens, the attention remains locally biased: query tokens disproportionately attend to patches within the same window while underweighting semantically similar tokens outside the window. This is attributed to the fact that query embeddings are constructed solely from inner-window features, limiting their global awareness. To alleviate this issue, we derive proxy tokens to serve as globally representative queries by aggregating the patches highly similar to the given query across the entire image. These proxies help reduce local bias and encourage semantically aligned attention across windows. Finally, we introduce a dynamic attention normalization strategy that adjusts attention magnitudes based on object scale, estimated from each query's similarity distribution. This adaptation ensures that small objects are not overwhelmed by irrelevant global tokens, while large objects benefit from richer contextual clues. Our results across multiple benchmarks demonstrate the effectiveness of our proposed approaches in mitigating the semantic discrepancy between different sliding windows.

Our contributions are summarized as follows:

- To our knowledge, this is the first work to identify and address the challenge stemming from the sliding window: locally bounded semantics in each window incur inconsistent prediction between windows.
- We extend the set of key-value tokens to cover the entire image, allowing each query to reason over global context regardless of its local window.
- We introduce globally representative proxy queries by aggregating high confidence tokens, enabling semantically consistent attention across windows.
- We propose a dynamic attention normalization mechanism that adapts the influence of global tokens based on the estimated object scale, improving robustness across varying object sizes and domains.
- Our framework can expand the receptive field of any baseline model without additional training.

## 2. Related Work

### 2.1. Open-Vocabulary Semantic Segmentation

The emergence of pre-trained vision-language models [19, 34] enables aligning text and image features for open-vocabulary semantic segmentation [29, 35]. Early methods follow a two-stage pipeline, where class-agnostic masks are first generated and then matched with text embeddings using models like CLIP [14, 15, 27, 30, 48]. While these methods are effective, they depend on external mask generators that are separate from the vision-language model. This separation can reduce visual-semantic consistency. To address this, single-stage methods [49, 51, 54] have been introduced. These methods directly predict segmentation masks from CLIP features and often use techniques such as attention bias [49] or mask pooling [51]. Other approaches include ODISE [47], which uses frozen diffusion features with a Mask2Former decoder [9], and CAT-Seg [11], which improves alignment by pixel-text similarity and text encoder fine-tuning. More recently, EBSeg [37] enhances generalization by adjusting image-text similarity distribution during training. Meanwhile, training-free approaches have gained attention for utilizing CLIP without requiring further training.

### 2.2. Training-Free OVSS

Training-free open-vocabulary semantic segmentation makes predictions without any additional training or fine-tuning. Instead of introducing learnable parameters, these approaches enhance the segmentation ability of CLIP by modifying its image encoder [26, 43] and leveraging vision foundation models [21, 24]. MaskCLIP [53] enhances spatial sensitivity in CLIP by discarding the query and key embeddings in the final self-attention block, using only the value embeddings. Subsequent methods have modified the attention mechanism by replacing query-key attention with value-value or self-self attention (e.g., query-query, key-key) [3, 26, 38, 42]. In addition, some approaches remove the final feed-forward layer and residual connections, as these components have been found to introduce

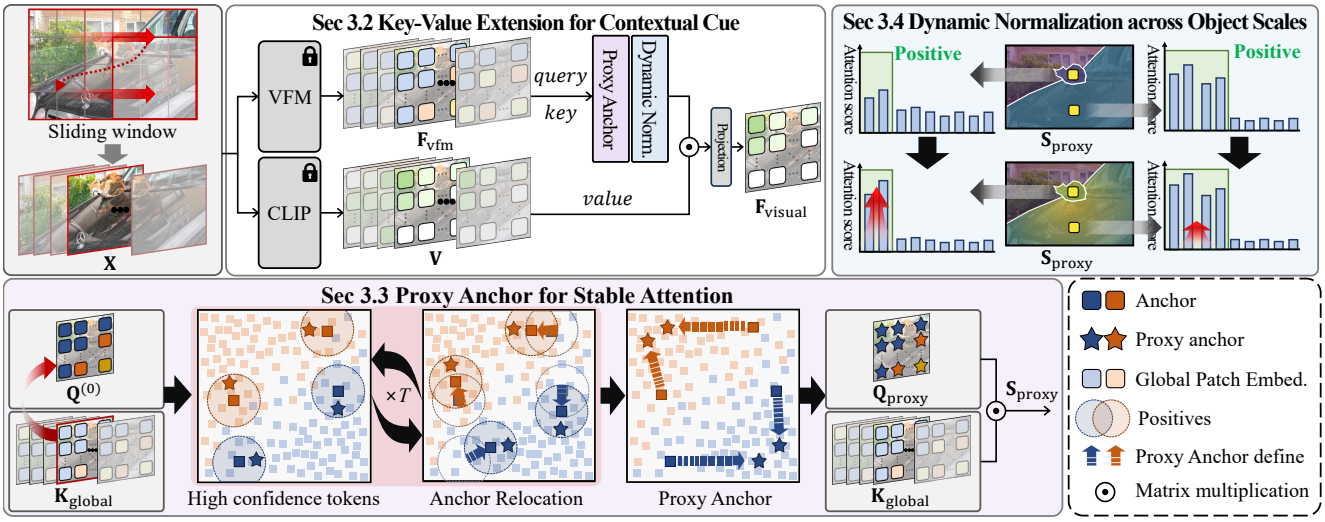



Figure 2. Overview of our proposed framework. The input image is first divided into overlapping windows and processed using frozen backbones: a Vision Foundation Model (VFM, e.g., DINO) and CLIP. We introduce a Key-Value Token Extension, where VFM features from the current window $\mathbf{F}_{\text{vfm}}$ serve as *query* tokens, while *key* tokens are gathered from all windows to provide global context. The corresponding *value* tokens $V$ are extracted from the final transformer layer of CLIP. Cross-attention is then applied, followed by a projection layer to generate the final visual features $\mathbf{F}_{\text{visual}}$. To stabilize attention across windows, each query token is replaced with a semantically representative proxy anchor. Finally, a dynamic normalization scheme adjusts attention strength based on object size, approximated by the number of positive samples associated with each proxy anchor.

noise in the resulting segmentation masks [18, 23]. ProxyCLIP [24] introduces a proxy attention mechanism that enhances spatial and semantic alignment in CLIP. However, most methods rely on sliding-window inference due to low training resolution, which makes it difficult for the model to understand the entire image and often leads to inconsistent results across different windows. To address these challenges, our method enhances global context and consistency by introducing key-value token extension and proxy-similarity. In doing so, it effectively mitigates attention imbalance and overcomes the inherent limitations of sliding-window inference.

# 3. Method

## 3.1. Preliminary

### 3.1.1. Open-Vocabulary Semantic Segmentation (OVSS)

OVSS leverages the visual-language alignment of CLIP [34]. Given an image $\mathbf{X}$, the CLIP visual encoder converts it as visual tokens $\mathbf{F}_{\text{visual}} \in \mathbb{R}^{N \times D}$, where $N$ is the number of tokens and $D$ is dimension size. Meanwhile, the text encoder processes a set of text prompts to project them to a text tokens $\mathbf{F}_{\text{text}}$. The cosine similarity between visual and text tokens determines logits, eventually categorizing each visual token to the text corresponding to the highest logits.

Recently, ProxyCLIP [24] has shown notable advances in OVSS by leveraging Vision Foundation Model (VFM) features. We adopt ProxyCLIP as our baseline, which replaces the attention map in the final transformer block of CLIP with a self-similarity map computed from features of a VFM. Specifically, the self-similarity map $\mathbf{S}$ is defined as:

$$\mathbf{S} = \mathbf{F}_{\text{vfm}} \cdot \mathbf{F}_{\text{vfm}}^{\top} \in \mathbb{R}^{N \times N}, \tag{1}$$

where $(\cdot)$ indicate matrix multiplication and $\mathbf{F}_{\text{vfm}}$ denotes the visual features extracted from a VFM (e.g., DINO [6]), which serves as the *query* and *key* embeddings in the attention mechanism.

To better integrate the self-similarity map into CLIP for open-vocabulary segmentation, they modulate and threshold the similarity scores to integrate semantically positive tokens. This refinement suppresses irrelevant token interactions and enables the similarity map to function as an effective attention map for segmentation tasks:

$$\mathbf{A} = \gamma \left( \mathbf{S} - \frac{\beta}{N^2} \sum_{i,j} [\mathbf{S}]_{ij} \right), \tag{2}$$

$$\mathcal{M}_{ij} = \begin{cases} 0, & A_{ij} \geq 0 \\ -\infty, & A_{ij} < 0, \end{cases} \tag{3}$$

$$\mathbf{Attn} = \text{SoftMax}(\mathbf{A} + \mathcal{M}), \tag{4}$$

where $\beta$ and $\gamma$ are fixed shift and scale hyperparameters.

The refined attention map $\mathbf{Attn}$ is applied to the *value* tokens $\mathbf{V} \in \mathbb{R}^{N \times D}$ from the final transformer block of CLIP. The final output is obtained via a projected matrix multiplication, following prior findings that skip residual and feed-forward layers [23]:

$$\mathbf{F}_{\text{visual}} = \text{Proj}(\mathbf{Attn} \cdot \mathbf{V}), \tag{5}$$

where Proj denotes the projection layer from the last transformer block of CLIP.

#### 3.1.2. Sliding-window

Since CLIP is pre-trained with a fixed input resolution of $224 \times 224$ pixels, it cannot directly process high-resolution inputs. Conventionally, this limitation is addressed by applying an overlapped sliding-window strategy, where the input image is divided into $L$ sub-windows of CLIP-compatible size. We follow this approach and perform OVSS independently on each sub-window. For overlapping regions, the logits are averaged across windows to produce the final per-patch predictions.

### 3.2. Key-Value Extension for Contextual Cue

While the sliding-window inference strategy is effective for adapting CLIP to high-resolution inputs by preserving its pre-trained resolution, it inherently restricts the model's receptive field to local window regions. This constrained field of view prevents the model from accessing holistic scene-level context, thereby hindering its ability to aggregate semantically relevant cues across spatially distant regions. The limitation becomes especially pronounced when semantically coherent regions (such as large objects or spatially continuous stuff categories) are partitioned across multiple windows. Such contextual discontinuity often leads to inconsistent or erroneous semantic predictions, as shown in Fig. 1.

To enhance contextual awareness in attention computation, we propose a key-value token extension, which expands the attention scope of each window by allowing it to reference visual information from the entire image, not just its local region. Specifically, we collect the visual features extracted from the VFM to serve as global *key* embeddings, and simultaneously retrieve the corresponding *value* tokens from the final transformer layer of CLIP. By aggregating these tokens across all $L$ windows, we construct a unified set of global *key* $\mathbf{K}_{\text{global}}$ and *value* tokens $\mathbf{V}_{\text{global}}$ as follows:

$$\mathbf{K}_{\text{global}} = [\mathbf{F}^{(1)}_{\text{vfm}}; \mathbf{F}^{(2)}_{\text{vfm}}; \ldots; \mathbf{F}^{(L)}_{\text{vfm}}] \in \mathbb{R}^{(LN)\times D}, \quad (6)$$

$$\mathbf{V}_{\text{global}} = [\mathbf{V}^{(1)}; \mathbf{V}^{(2)}; \ldots; \mathbf{V}^{(L)}] \in \mathbb{R}^{(LN)\times D}. \quad (7)$$

With these global tokens, we perform cross-window attention by computing similarity between the local *query* tokens from the current window and the aggregated global *key* tokens, followed by weighted aggregation over the corresponding *value* tokens:

$$\mathbf{A}_{\text{ext}} = \mathbf{Q} \cdot \mathbf{K}^{\top}_{\text{global}} \in \mathbb{R}^{N\times(LN)}, \quad (8)$$

$$\mathbf{F}_{\text{visual}} = \text{Proj}(\mathbf{A}_{\text{ext}} \cdot \mathbf{V}_{\text{global}}) \in \mathbb{R}^{N\times D}. \quad (9)$$

Here, $\mathbf{Q} \in \mathbb{R}^{N\times D}$ denotes the set of query tokens extracted from the $\mathbf{F}_{\text{vfm}}$ of the current sliding window. These serve as the local anchors in the attention operation and initiate semantic integration with globally gathered keys and values.

### 3.3. Proxy Anchor for Stable Attention

After applying the key-value token extension, we observe a persistent locality bias: attention is predominantly concentrated on tokens within the same window (inner-window), while downweighting semantically similar tokens from other windows (outer-window), as shown in Fig. 3(a). Notably, even when outer-window tokens correspond to the same spatial location in the original image (i.e., overlapping regions), attention remains biased toward inner-window tokens. This behavior limits the model's capacity to aggregate consistent semantics across windows.

To alleviate this bias, we introduce a proxy-based attention mechanism that replaces each query token with a semantically representative proxy aggregated from tokens with high similarity across all windows. This proxy serves as a stable anchor that reflects the underlying semantics of the region, allowing attention to be allocated based on semantic coherence rather than window-local bias. Consequently, our method promotes more globally consistent attention patterns and enhances cross-window integration.

Specifically, for a given query token, we follow [36] to repeatedly compute the mean vector of high-similarity neighbors, gradually converging on a proxy that best represents the shared semantics around the anchor. We define the initial proxy-query $\mathbf{Q}^{(0)}_i$, which stems from $i$-th token of local query tokens $\mathbf{Q}$ and construct the index set of high-confident tokens $\mathcal{P}^{(0)}_i$, which indicates a higher cosine similarity compared to a pre-defined threshold $\rho$ as below:

$$\mathcal{P}^{(0)}_i = \left\{ j \,\middle|\, \mathbf{Q}^{(0)}_i \cdot \mathbf{K}_{\text{global},j} > \rho,\ j \in \mathcal{J} \right\}, \quad (10)$$

where $\mathcal{J}$ denotes the index set comprising all patch features of all windows. Also, the $t$-th proxy is defined as follows:

$$\mathbf{Q}^{(t)}_i = \frac{1}{\left|\mathcal{P}^{(t-1)}_i\right|} \sum_{j \in \mathcal{P}^{(t-1)}_i} \mathbf{K}_{\text{global},j}. \quad (11)$$

By repeating $T$ steps of Eq. (10) and (11) for generating refined proxy, we obtain $\mathbf{Q}^{(T)}_i$. This operation is computed for all *query* tokens. Using the proxy $\mathbf{Q}_{\text{proxy}} = \bigcup_i \mathbf{Q}^{(T)}_i$, proxy anchor-based attention map for the current window is represented as follows:

$$\mathbf{S}_{\text{proxy}} = \mathbf{Q}_{\text{proxy}} \cdot \mathbf{K}^{\top}_{\text{global}} \in \mathbb{R}^{N\times(BN)}. \quad (12)$$

The proxy is located at the center of the high-similarity embedding samples, aggregated from both inner- and outer-window regions. As a result, the proxy inherently balances the contributions of tokens across windows, allowing attention to be allocated more uniformly to semantically relevant regions, as shown in Fig. 3(b).

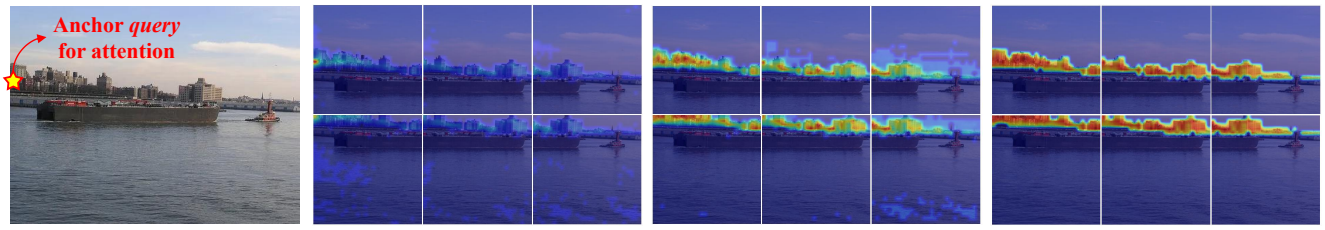



Image (a) Key-Value Extension (b) + Proxy-based Attention (c) + Dynamic Normalization

Figure 3. Visualization of attention maps for an anchor *query* token. Proxy-based attention enhances focus on semantically relevant regions across both inner- and outer-window areas. The subsequent dynamic normalization further suppresses irrelevant responses, especially from noisy tokens, yielding sharper and more semantically consistent attention distributions.

### 3.4. Dynamic Normalization across Object Scales

Our key-value extension allows each query to attend to tokens from global windows. While this enables broader contextual integration, it also increases the risk of attending to irrelevant or semantically mismatched tokens (i.e., negatives). This issue is strongly influenced by object scale: when a query corresponds to a small object or fine-grained region, the number of relevant positive tokens tends to be low. In such cases, attention is more easily dominated by irrelevant tokens, leading to inaccurate or noisy predictions.

To address this, we propose a dynamic normalization strategy that modulates and thresholds the similarity map in a scale-aware manner. The key idea is that when a query is associated with only a few positive tokens, as is often the case with small objects, the attention becomes more prone to being dominated by irrelevant or negative tokens, leading to degraded prediction quality. Our method adaptively tightens the normalization in such cases to suppress the effect of these irrelevant signals. Conversely, when a query corresponds to a large object and is surrounded by many positive tokens, the normalization becomes more permissive to enhance their relative influence in the attention computation. Additionally, the influence of irrelevant tokens increases with the number of outer-window tokens, amplifying noise in the attention computation. To handle this more effectively, our normalization strategy adjusts its scaling behavior based on the number of windows, suppressing the impact of such noise under larger global contexts.

Specifically, we introduce two adaptive variables, $\mathbf{u}$ and $\mathbf{w}$, as adaptive counterparts to the fixed hyperparameters in ProxyCLIP [24] to modulate normalization accordingly. The attention score for the $i$-th query anchor is computed as:

$$\mathbf{Attn}_i = \mathbf{w}_i \left( \mathbf{S}_{\text{proxy}} - \frac{\mathbf{u}}{NL} \sum_{j=1,\dots,NL} [\mathbf{S}_{\text{proxy}}]_{ij} \right), \quad (13)$$

where $\mathbf{u}$ is a shifting variable dependent on the number of windows $L$, controlling the overall influence of extended tokens. $\mathbf{u}$ is defined as:

$$\mathbf{u} = 1 + \lambda_1 \log(1 + L), \quad (14)$$

where is a fixed coefficient. As $L$ increases, $\mathbf{u}$ grows, resulting in more conservative normalization to suppress noisy token influence. Next, we define the scaling variable $\mathbf{w}_i$ based on the number of high confidence tokens $|\mathcal{P}_i|$ associated with each *query*:

$$\mathbf{w}_i = 1 + \frac{\lambda_2}{|\mathcal{P}_i|}. \quad (15)$$

A smaller $|\mathcal{P}_i|$, often observed in small objects, leads to stronger amplification of the most relevant tokens while suppress the irrelevant tokens, encouraging scale-aware attention allocation. Note that, $\lambda_1$ and $\lambda_2$ are shared across all datasets. Finally, replacing $\mathbf{A}$ in Eq. (9) with $\mathbf{Attn}$ in Eq. (13) yields the final visual tokens for prediction. The masking and softmax operations are then applied, as formulated in Eq. (4).

A key distinction of our method is a per-query normalization strategy, compared to the prior approach [24] that applies identical normalization across all windows and tokens. By adapting attention modulation at the level of each anchor token, our formulation enables fine-grained control over the influence of global context. This scale-aware behavior is particularly effective for balancing attention across varying object sizes. Furthermore, Dynamic Normalization copes with object scale variations, which differ among datasets, and thus removes the necessity of dataset-specific hyperparameters unlike recent methods [8, 22, 50].

## 4. Experiment

### 4.1. Experimental Settings

**Datasets and Evaluation Metrics** We evaluate the proposed model on eight semantic segmentation benchmarks. We note that five datasets (Pascal VOC21 [17], Pascal Context60 [32], COCO-Object [4], ADE20K [52], and Cityscapes [12]) include an explicit background class, while other datasets (Pascal VOC20 [17], Pascal Context59 [32], and COCO-Stuff 164K [4]) only contain object classes. In the same order, they comprise 21, 60, 81, 150, 19, 20, 59, and 171 semantic classes, respectively. Following standard practice, we report performance using mean Intersection-over-Union (mIoU).

| Model | Train | External Model | *DS Hyperparameter | With Background V21 | PC60 | C-Obj | V20 | Without Background PC59 | C-Stf | City | ADE | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CLIP-DINOiser Setting** | 448 Image Resize, 448 Crop size, 224 Stride, Without rename trick† | | | | | | | | | | | |
| GroupViT [46] | ✓ | - | ✘ | 50.4 | 18.7 | 27.5 | 79.7 | 23.4 | 15.3 | 11.1 | 9.2 | 29.4 |
| ReCo [40] | ✓ | - | ✘ | 25.1 | 19.9 | 15.7 | 57.7 | 22.3 | 14.8 | 21.6 | 11.2 | 23.5 |
| CLIP-DIY [44] | ✘ | - | ✘ | 59.9 | 19.7 | 31.0 | 79.7 | 19.8 | 13.3 | 11.6 | 9.9 | 30.6 |
| TCL [7] | ✓ | - | ✘ | 55.0 | 30.4 | 31.6 | **83.2** | 33.9 | 22.4 | 24.0 | 17.1 | 37.2 |
| CLIP-DINOiser [45] | ✓ | DINO | ✘ | 62.2 | 32.4 | 35.0 | 80.2 | 35.9 | 24.6 | 31.7 | **20.0** | 40.3 |
| **ProxyCLIP + GLA**‡ | ✘ | DINO | ✘ | **63.4** | **35.8** | **37.3** | 80.4 | **39.4** | **26.3** | **33.5** | 19.1 | **41.9** |
| **ClearCLIP Setting** | 448 Image Resize, 448 Crop size, 224 Stride, Without rename trick | | | | | | | | | | | |
| ClearCLIP [23] | ✘ | - | ✘ | 51.8 | 32.6 | 33.0 | **80.9** | 35.9 | 23.9 | 30.0 | 16.7 | 38.1 |
| **ClearCLIP + GLA** | ✘ | - | ✘ | **55.6** | **33.2** | **34.6** | 79.7 | **36.5** | **24.6** | **32.7** | **17.4** | **39.3** |
| **ProxyCLIP Setting** | 448 Image Resize, 336 Crop size, 112 Stride, With rename trick on Background Classes | | | | | | | | | | | |
| ProxyCLIP [24] | ✘ | DINO | ✘ | 61.3 | 35.3 | 37.5 | 80.3 | 39.1 | 26.5 | 38.1 | 20.2 | 42.3 |
| **ProxyCLIP + GLA** | ✘ | DINO | ✘ | **63.2** | **35.8** | **37.7** | **81.5** | **39.7** | **26.8** | **38.5** | **20.3** | **42.9** |
| **SCLIP Setting** | 336 Image Resize, 224 Crop size, 112 Stride, With rename trick | | | | | | | | | | | |
| SCLIP [42] | ✘ | - | ✘ | 59.1 | 30.4 | 30.5 | 80.4 | 34.2 | 22.4 | 32.2 | 16.1 | 38.2 |
| **SCLIP + GLA** | ✘ | - | ✘ | **59.8** | **32.0** | **32.8** | **80.5** | **35.1** | **24.2** | **35.1** | **18.0** | **39.8** |
| LaVG [20] | ✘ | - | ✘ | 62.1 | 31.6 | 34.2 | 82.5 | 34.7 | 23.2 | 26.2 | 15.8 | 38.8 |
| NACLIP [18] | ✘ | - | ✘ | 58.9 | 32.2 | 33.2 | 79.7 | 35.2 | 23.3 | 35.5 | 17.4 | 39.4 |
| ResCLIP [50] | ✘ | - | ✓ | 61.1 | 33.5 | 35.0 | 86.0 | 36.8 | 24.7 | 35.9 | 18.0 | 41.4 |
| FreeCP [8] | ✘ | - | ✓ | 65.8 | 35.3 | 37.2 | 84.3 | 38.0 | 24.9 | 33.3 | 18.4 | 42.2 |
| DIH-CLIP [16] | ✘ | - | ✘ | 64.2 | 36.0 | 37.4 | 84.9 | 39.7 | 24.5 | 40.2 | 19.6 | 43.3 |
| FLOSS [2] | ✘ | - | ✘ | - | - | - | 80.2 | 35.9 | 23.6 | 37.0 | 18.4 | - |
| CASS [22] | ✘ | DINO | ✓ | 65.8 | **36.7** | **37.8** | **87.8** | **40.2** | 26.7 | 39.4 | 20.4 | **44.4** |
| **ProxyCLIP + GLA (Best**§**)** | ✘ | DINO | ✓ | **66.7** | 36.3 | 37.7 | 84.7 | **40.2** | **27.2** | **41.2** | **20.5** | 44.3 |
| **ProxyCLIP + GLA** | ✘ | DINO | ✘ | 66.3 | 36.1 | 37.7 | 84.2 | 39.9 | 26.9 | 40.8 | 20.0 | 44.0 |

Table 1. Open-vocabulary semantic segmentation results on 8 datasets using CLIP ViT-B/16. *DS Hyperparameter: dataset-specific hyperparameters (not used in Ours). †Rename trick: multiple textual prompts per class. ‡GLA: Adaptation for ClearCLIP [23], SCLIP [42], and ProxyCLIP [24]. § Best: optionally uses dataset-specific hyperparameters (**u**, **w**), by manual tuning.

**Implementation Details.** Our settings follow existing training-free open-vocabulary semantic segmentation models [23, 24, 42, 45], including text prompt policy, background threshold, image resizing, sliding window crop size, and stride (details in the Appendix B). We use frozen CLIP (ViT-B/16) to extract features and DINO (ViT-B/8) as VFM, similar to ProxyCLIP [24]. Meanwhile, the query and key tokens from CLIP can also be utilized to form attention maps, replacing VFM features. ClearCLIP [23] and SCLIP [42] adopt this strategy, and our framework can similarly support it. However, in this case, a separate query smoothing step and hyperaprater tuning is necessary, as detailed in the Appendix C. All experiments are conducted on a single NVIDIA Titan RTX GPU. Proxy-anchor hyperparameters are set to $\rho = 0.6$ with 2 proxy generation steps, and Dynamic Normalization uses $\lambda_1 = 0.3$ and $\lambda_2 = 30$.

## 4.2. Comparison with State-of-the-Art Methods

**Quantitative Results.** As shown in Tab. 1, we compare our method with existing open-vocabulary semantic segmentation (OVSS) approaches and report its performance under each setting for a fair evaluation. Our method achieves an average mIoU gain of 1.6% under the CLIP-DINOiser setting, demonstrating clear advantages over training-based models. When integrated with ClearCLIP [23], ProxyCLIP [24], and SCLIP [42], it further yields improvements of 1.2%, 0.6%, and 1.6%, respectively, highlighting its adaptability on any baseline model. In contrast to recent training-free OVSS methods [8, 10, 22, 50], which depend on dataset-specific hyperparameters to enhance performance at the cost of generality, our approach attains an average mIoU of 44.0%, outperforming existing models without tuning. When dataset-specific hyperparameters are optionally applied, our model achieves even higher scores, comparable to all baseline methods. These results indicate that our approach is both effective and generalizable, making it readily applicable across diverse datasets.

**Qualitative Results.** Fig. 4 shows qualitative comparisons with recent state-of-the-art methods on Pascal VOC21 [17], COCO-Stuff164k [4], and Cityscapes [12]. Our method generates cleaner segmentation maps with fewer misclassified pixels and effectively suppresses the grid-like artifacts commonly seen in ProxyCLIP [24] and CASS [22]. By leveraging contextual cues from global windows, our model produces more consistent predictions and preserves clear object boundaries. For example, in the rightmost scene in Cityscapes, ProxyCLIP misclassifies the road center as a sidewalk, whereas ours accurately distinguishes the road and sidewalk regions without noisy or inconsistent outputs.

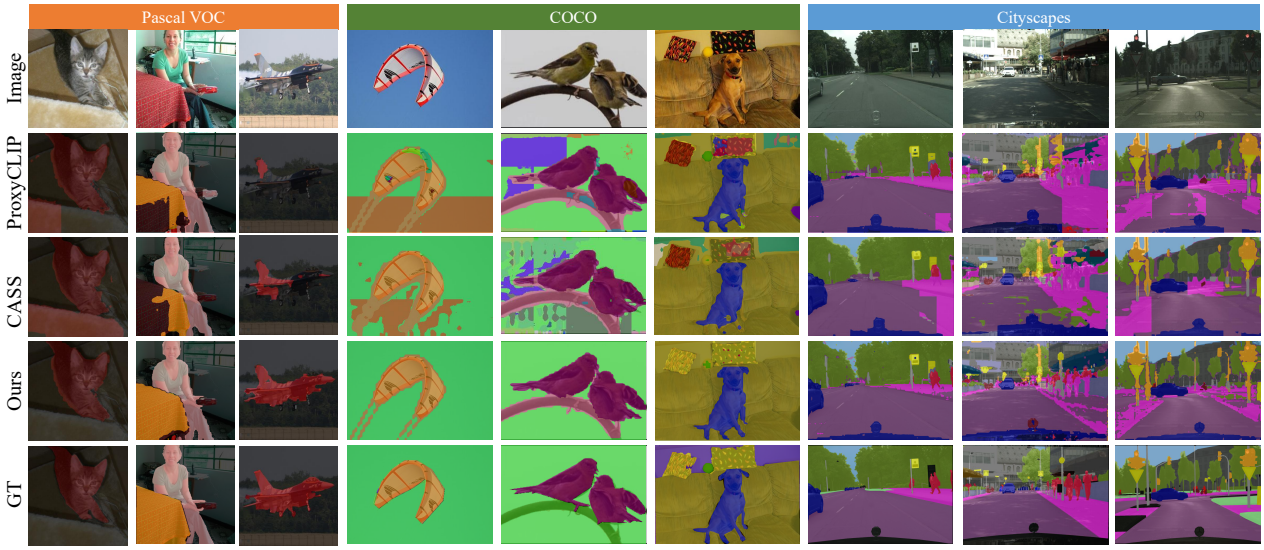



Figure 4. Qualitative results among ProxyCLIP [24], CASS [22], Ours on Pascal VOC21 [17], COCOstuff [4], and Cityscapes [12].

| | KVE | Proxy | Norm | | | mIoU |
|---|---|---|---|---|---|---|
| | | | Fixed | Best | Dyna | |
| (a) | | | | | | 30.8 |
| (b) | ✓ | | | | ✓ | 43.1 |
| (c) | | ✓ | | | ✓ | 43.0 |
| (d) | ✓ | ✓ | ✓ | ✓ | | 44.3 |
| (e) | ✓ | ✓ | | ✓ | ✓ | 44.0 |

Table 2. Ablation study of each component in our method. KVE: Key-Value Extension, Proxy: Proxy Anchor-based Attention, Norm: Normalization (Best, Dynamic). Dataset-specific tuning is applied for 'Best' normalization configuration, with detailed results in Tab. 1.

# 5. Further Analysis

## 5.1. Ablation Study

To validate the effectiveness of each component, we conduct an ablation study based on five experimental configurations, as summarized in Tab. 2. We begin with a baseline (a) that computes attention maps using features from DINO, restricted to inner-window tokens only, without any form of normalization. Comparing (a) and (b), we observe a clear performance improvement when applying key-value extension along with dynamic normalization. This demonstrates the importance of incorporating global context and stabilizing the attention computation. Next, comparing (b) and (e), the introduction of proxy-based attention in (e) leads to an additional performance gain. This improvement stems from the proxy's ability to produce more semantically grounded and stable attention distributions, especially in outer-window regions. We also compare (c) and (e) to isolate the contribution of the key-value extension. The results show that incorporating key-value tokens from outer windows yields a performance improvement, confirming that these extended tokens provide valuable contextual cues for disambiguating local predictions. Finally, experiment (d) reports results using dataset-specific hyperparameters, which were searched to maximize performance for each dataset. When comparing (d) and our method (e), we find that the performance gap is minimal, indicating that our adaptive normalization approach effectively generalizes across datasets without the need for manual tuning.

## 5.2. Effect of Attention Stabilization

We visualize the effect of each method on attention maps in Fig. 3. In Fig. 3(a), the attention maps after applying Key-Value Extension show that attention initially concentrates on a few tokens within the inner-window region, indicating that the model struggles to distribute attention across object boundaries and capture object-level semantics beyond individual windows. Over time, the strong attention weights originally confined to inner-window positives gradually spread to outer regions, leading to a holistic, object-level attention allocation. In Fig. 3(b), after applying Proxy-Attention, attention extends across the entire object regardless of window boundaries, showing that proxy queries effectively stabilize attention. Subsequently, in Fig. 3(c), introducing Dynamic Normalization suppresses irrelevant tokens and focuses attention on meaningful object regions, highlighting its role in refining attention distribution.

## 5.3. Object Scale via high confidence Tokens

We conduct experiments to validate the assumption that high confidence tokens are correlated with object scale in Dynamic Normalization. During proxy construction, we use the number of high confidence tokens, $|\mathcal{P}_i|$, as an estimate of object scale and incorporate it into the attention scaling factor, as defined in Eq. 15. In Fig. 5, we analyze whether the number of high confidence tokens aligns with the ground-truth object scale on the Cityscapes [12], which contains objects of diverse sizes. After key-value

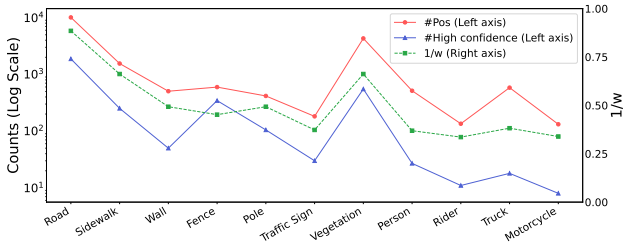



Figure 5. Class-wise object scale and 1/**w** in Cityscapes

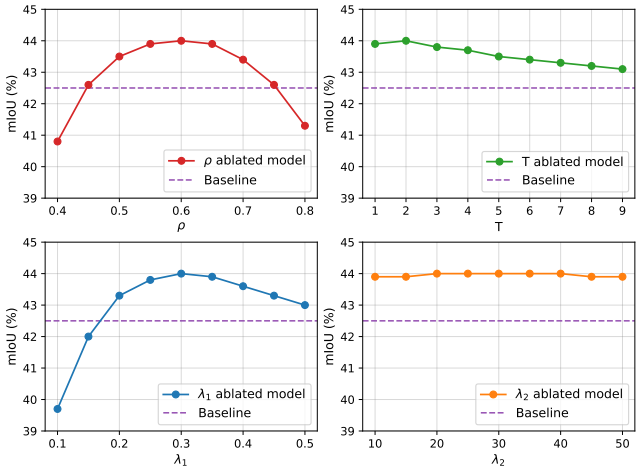



Figure 6. Hyperparameter sensitivity experiment.

extension, we measure both the number of patch tokens (#Pos) and high confidence tokens (#High confidence) for each class per image, and report the class-wise average attention scaling factor **w**, which is inversely proportional to #High confidence. We observe consistent trends across all three metrics: large objects (e.g., Road and Terrain) exhibit a greater number of high confidence tokens and correspondingly lower attention scaling factors, whereas small objects (e.g., Person and Rider) contain fewer tokens and thus receive higher attention scaling.

### 5.4. Hyperparameter Sensitivity

We leverage four hyperparameters, $\rho$, T, $\lambda_1$, and $\lambda_2$, and conduct a sensitivity analysis to examine their effect (Fig. 6). Firstly, $\rho$ controls the selection of high confidence tokens in the cosine similarity scale. If $\rho$ is too low, negatives may be fused with positives, causing semantic collapse in proxy anchors and degrading performance, while too high $\rho$ prevents proxies from aggregating sufficient positives, also reducing performance. The repetition step T in proxy anchor construction slightly affects performance, as a higher T can lead to semantic collapse. In Dynamic Normalization, $\lambda_1$ determines the attention masking threshold. A low $\lambda_1$ lowers the threshold, causing fewer tokens to be masked and more dispersed attention, whereas a higher $\lambda_1$ increases masking and helps to focus attention effectively. In contrast, $\lambda_2$ has minimal impact on performance, revealing its robustness. Overall, all hyperparameters exhibit a broad range to consistently outperform the baseline model.

| CLIP | Model | V21 | PC59 | C-Stf | Avg. |
|---|---|---|---|---|---|
| ViT-B/32 | ProxyCLIP | 57.3 | 35.2 | 23.6 | 38.7 |
| | CASS | 58.2 | 36.5 | 24.4 | 39.7 |
| | Ours | **61.8** | **37.1** | **25.2** | **41.4** |
| ViT-L/14 | ProxyCLIP | 60.7 | 38.2 | 26.2 | 41.7 |
| | CASS | 62.1 | **39.1** | 26.3 | 42.5 |
| | Ours | **63.5** | **39.1** | **26.7** | **43.1** |

Table 3. Segmentation results of different CLIP backbones.

| VFM | Model | V21 | PC59 | C-Stf | Avg. |
|---|---|---|---|---|---|
| DINOv2 [33] | ProxyCLIP | 59.6 | 37.3 | 25.3 | 40.7 |
| | +GLA | **60.8** | **37.8** | **25.7** | **41.4** |
| DINOv3 [41] | ProxyCLIP | 62.8 | 38.8 | 26.1 | 42.6 |
| | +GLA | **63.6** | **38.9** | **26.3** | **42.9** |

Table 4. Segmentation results of different VFM backbones.

### 5.5. Generalization Across Backbones

To assess the generalizability of our framework across varying CLIP backbone capacities, we evaluate performance using ViT-B/32 and ViT-L/14. As shown in Tab. 3, ProxyCLIP and CASS exhibits a substantial performance drop of 4.4%, 4.5% in average mIoU when switching to the lower capacity ViT-B/32. In contrast, our method incurs only a 3.0% decrease, demonstrating improved robustness to representational capacity degradation. This indicates that our model more effectively leverages the available features across diverse CLIP architectures, and is less susceptible to variations in visual encoder strength. Furthermore, in Fig. 4, we conduct backbone exchanges in VFM to demonstrate the adaptability of our module. Specifically, we replace the backbone from DINO ViT-B/16 to DINOv2-reg ViT-L/14 [13, 33] and DINOv3 ViT-B/16 [41] to verify the generality of our method. In each case, our approach yields performance improvements of +0.7% and +0.3%, respectively. Additionally, our method is compatible with complementary techniques such as multi-layer feature fusion and text-prompt modifications, as discussed in the Appendix H, I. Owing to its generality, our framework can be readily integrated with new backbones and future improvements.

## 6. Conclusion & Limitation

In this paper, we propose GLA-CLIP, a training-free method for OVSS that resolves inconsistent predictions across sliding windows. Using Key-Value Extension, each query attends to globally aggregated tokens, mitigating semantic discrepancies. We further address window bias and small object neglect by Proxy Anchor and Dynamic Normalization. While the Key–Value Extension utilizes whole tokens from other windows, computational cost may increase. More research is needed to efficiently leverage high-quality tokens across different windows. While our method can be adopted on any backbone features, certain hyperparameters may need to be adjusted to achieve optimal performance. We hope future research will address both memory efficiency and hyperparameter generalization.

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